

100 Days of Azure DevOps

Day 2 Handout — Portal is the map; CLI is GPS; PowerShell is the object knife

1 - Azure & DevOps Foundations · Personal lab only · Educational content · Not a sales pitch

Architecture A — three clients, one ARM control plane

Hop	What it is	Proof in the lab
You	Intent: create RG, list, delete	Decide rg-day02-lab / centralindia
Portal	portal.azure.com ARM JSON via the blade	Create RG once with clicks to feel the cost
Azure CLI	az → REST to management.azure.com	az group create --name rg-day02-lab
Azure PowerShell	Az module cmdlets, PS objects	New-AzResourceGroup when you need objects
ARM	Auth, RBAC, policy, lock, provider routing	Same 201/202 whether click or CLI
Resource provider	Microsoft.Resources/resourceGroups	az group show -n rg-day02-lab -o jsonc

Architecture B — pick the tool by the job, not by comfort

Job	Default	Move off that default when
First look at a service	Portal	You already know the ARM type and SKU
Same task twice this week	az script in a gist / repo	You are still clicking the same blade
Put it in a pipeline later	AzureCLI@2 or az in YAML	Classic click-ops cannot be reviewed
Need objects / Windows estate	PowerShell Az	JSON text from az is enough
Naming	rg-, st, app-, pip- prefixes	You ship 'New Resource Group (1)'
Output	az -o table for humans, jsonc for diffs	You grep unstructured Portal screenshots

Architecture C — pitfalls that silently target the wrong subscription

Pitfall	What you see	Fix
az login to work tenant	az account show lists employer sub	az logout; az login; az account set --subscription <personal-id>
Cloud Shell vs local CLI	Commands work in Portal shell, fail on laptop	Install https://aka.ms/installazurecliwindows ; az version
Default location drift	RG in eastus because last session stuck	Always pass --location / -l on create
Delete in Portal, create in CLI	Two RGs, two names, two bills	One name: rg-day02-lab; az group delete --yes --no-wait
PowerShell 5 vs 7 mix	Connect-AzAccount missing Az.Accounts	Install-Module Az -Scope CurrentUser
json vs table	You cannot see properties you need	az group show -o jsonc then --query

One-liner to remember

Explore once in Portal. Second time use CLI. Forever after, script it.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal Microsoft account and personal Azure subscription only. Install CLI on your own PC.

- 1 Install Azure CLI from <https://aka.ms/installazurecliwindows> (or brew/apt on your OS). Run az version.
- 2 Terminal: az login → pick the personal subscription. az account set --subscription "<personal-id>".
- 3 az group create --name rg-day02-lab --location centralindia. Confirm with az group show --name rg-day02-lab -o jsonc.
- 4 portal.azure.com → Resource groups → Create the same shape once (name rg-day02-portal-compare) to count the clicks.
- 5 az group list --output table. Compare properties: location, provisioningState, tags.
- 6 az group delete --name rg-day02-portal-compare --yes --no-wait. Keep or delete rg-day02-lab before Day 3.
- 7 Optional: pwsh → Connect-AzAccount → Get-AzResourceGroup -Name rg-day02-lab. Same ARM object, different client.

Login, pin the personal sub, create and inspect the RG

```
az login
az account set --subscription "<your-personal-subscription-id>"
az group create --name rg-day02-lab --location centralindia
az group show --name rg-day02-lab --output jsonc
az group list --query "[?starts_with(name, 'rg-day0')].{name:name, loc:location, state:properties.provisioningState}" -o table
az group delete --name rg-day02-lab --yes --no-wait
```

Done checklist

- Can explain Portal vs CLI vs PowerShell in one line each
- az login targets the personal subscription (az account show)
- Created rg-day02-lab via CLI and inspected JSON
- Deleted the compare RG (or documented why it remains)

Tomorrow — Day 3

Azure Resource Manager (ARM) basics